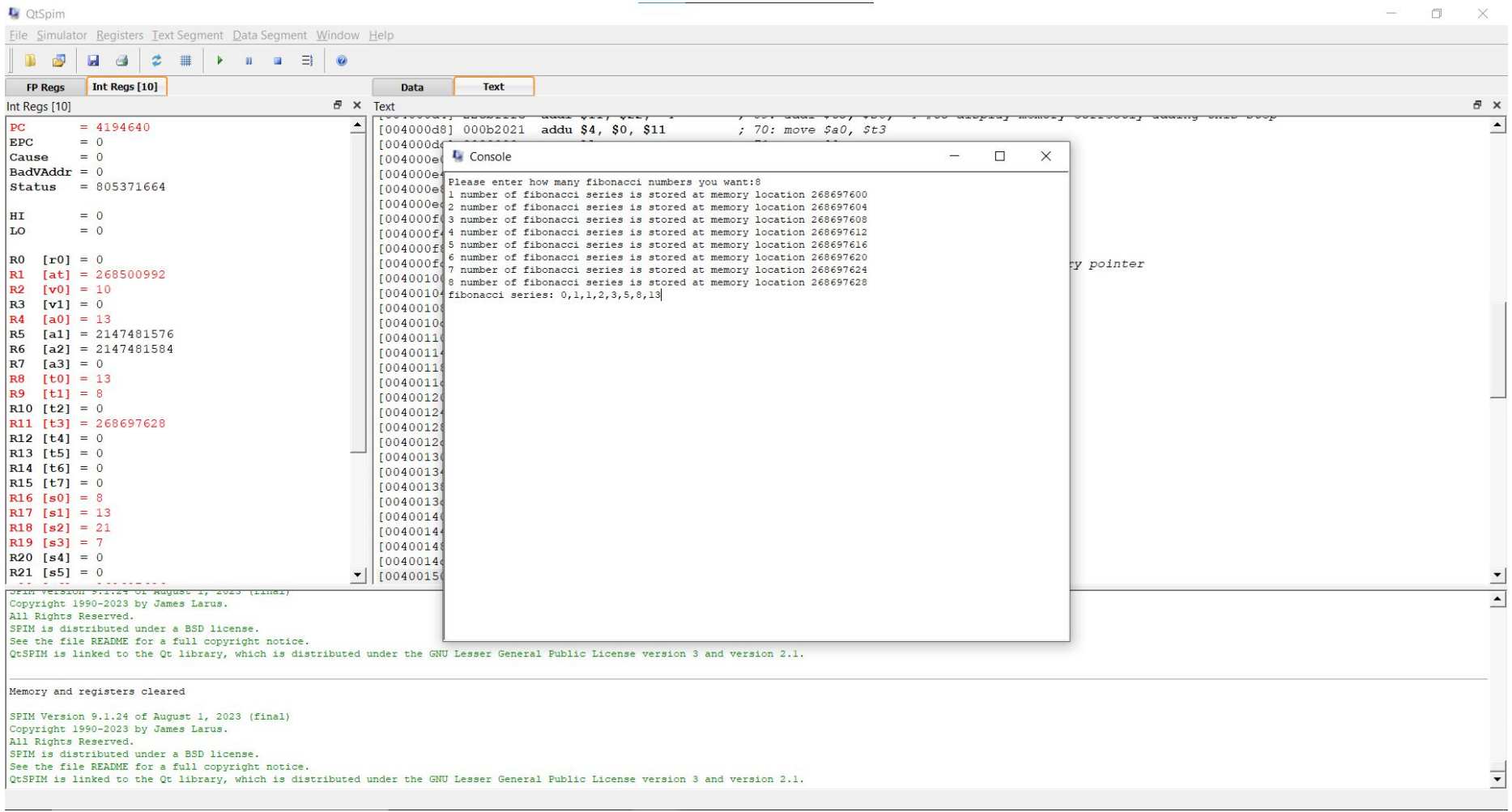
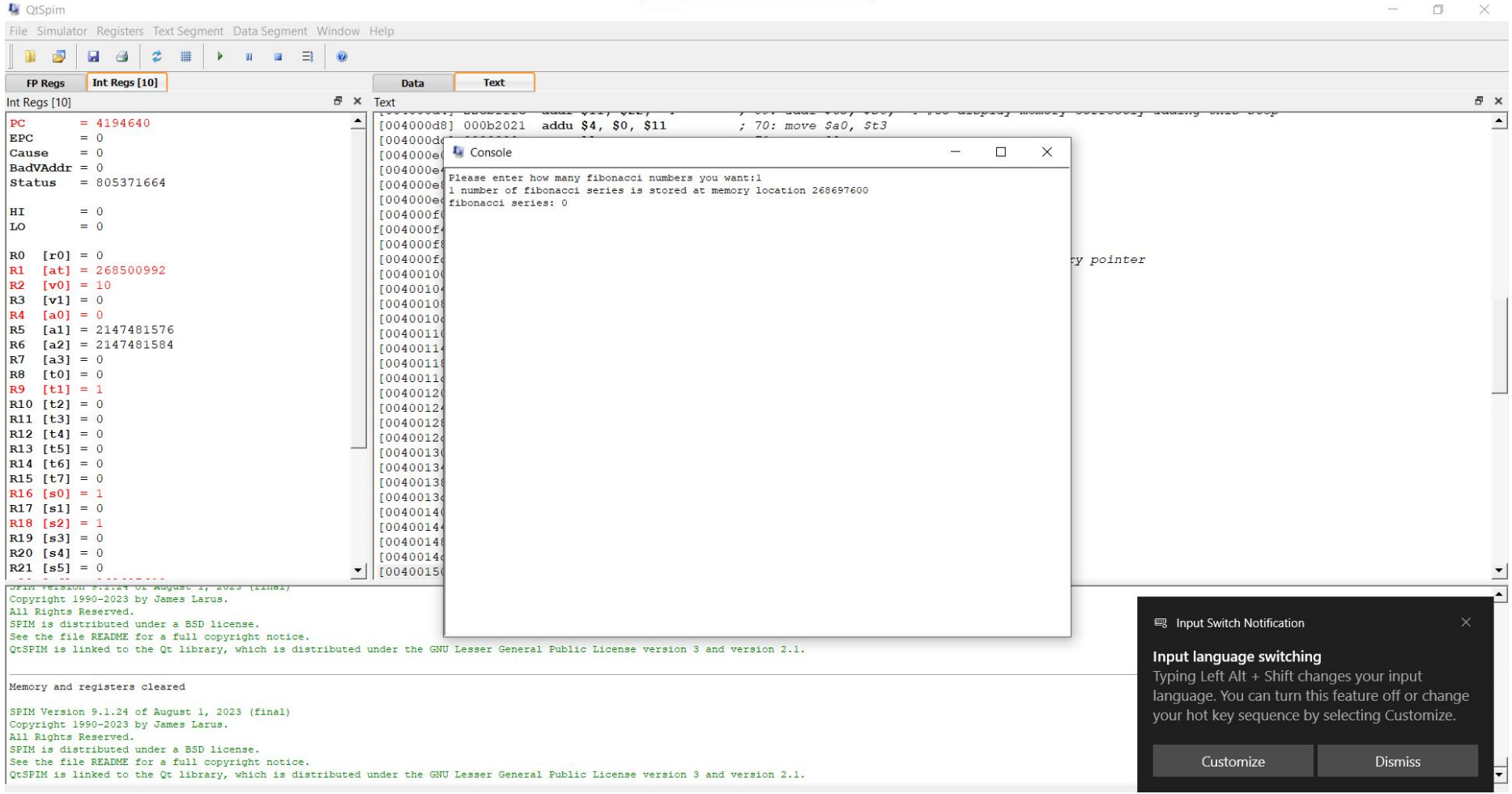
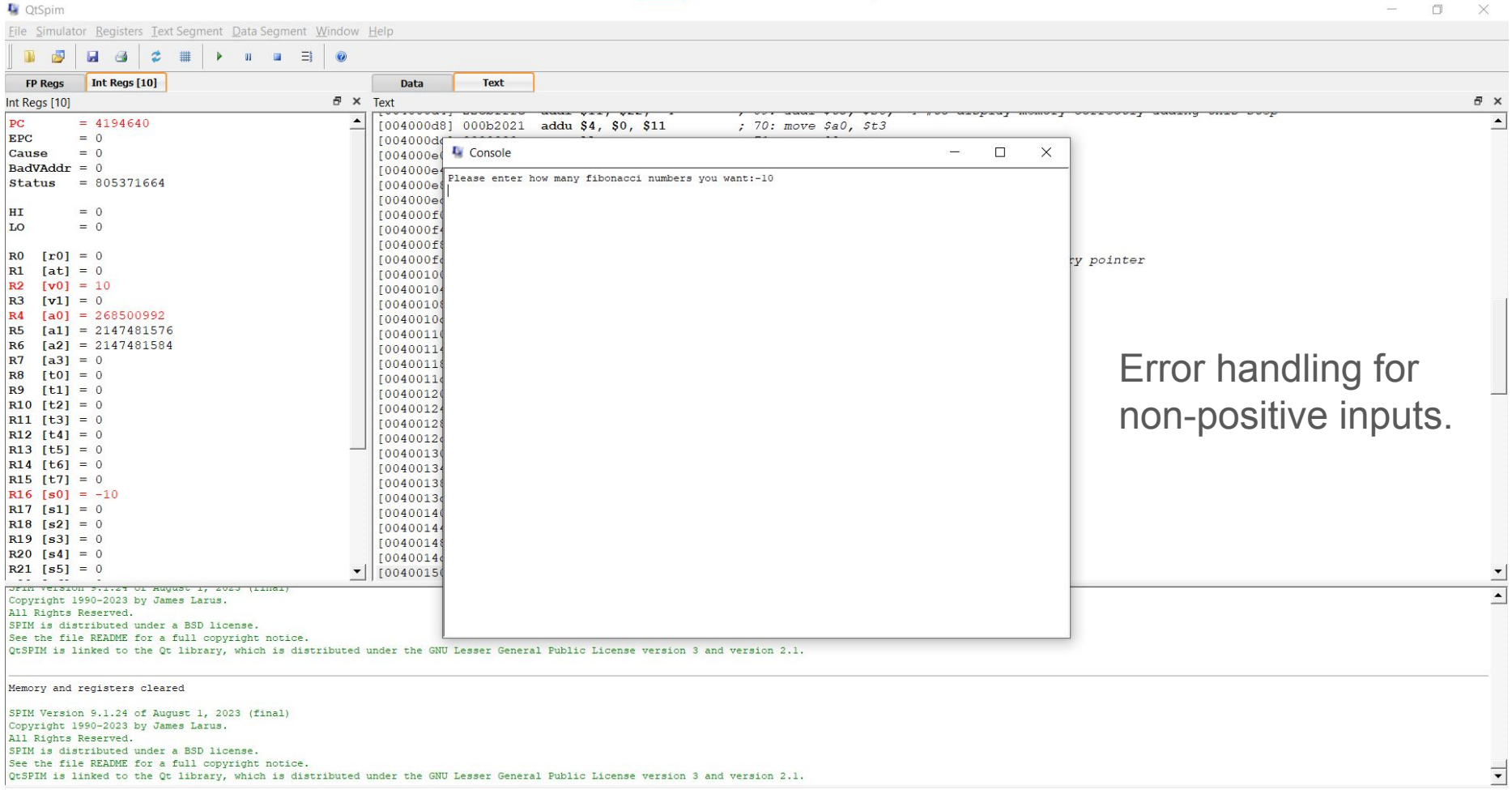
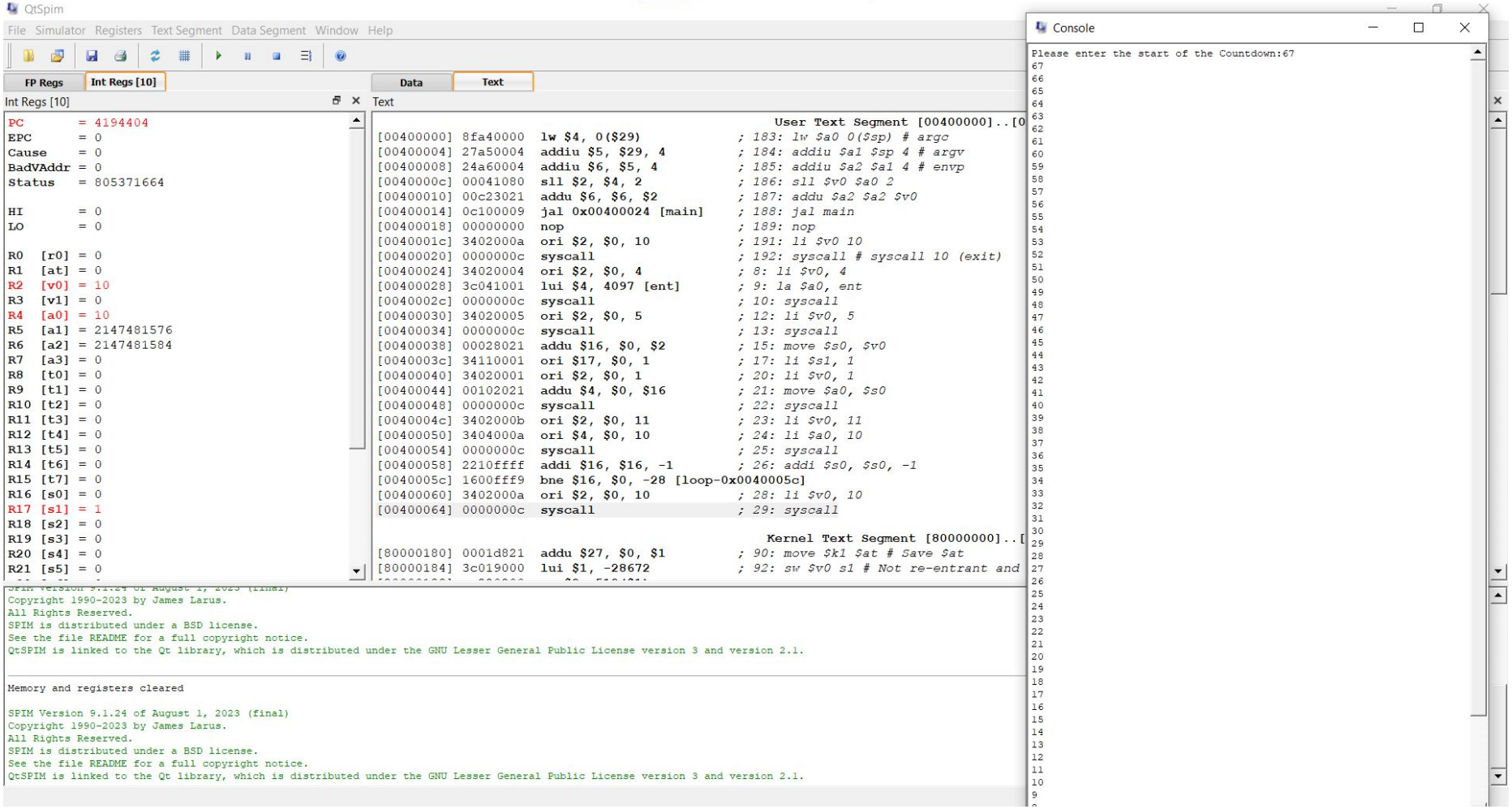
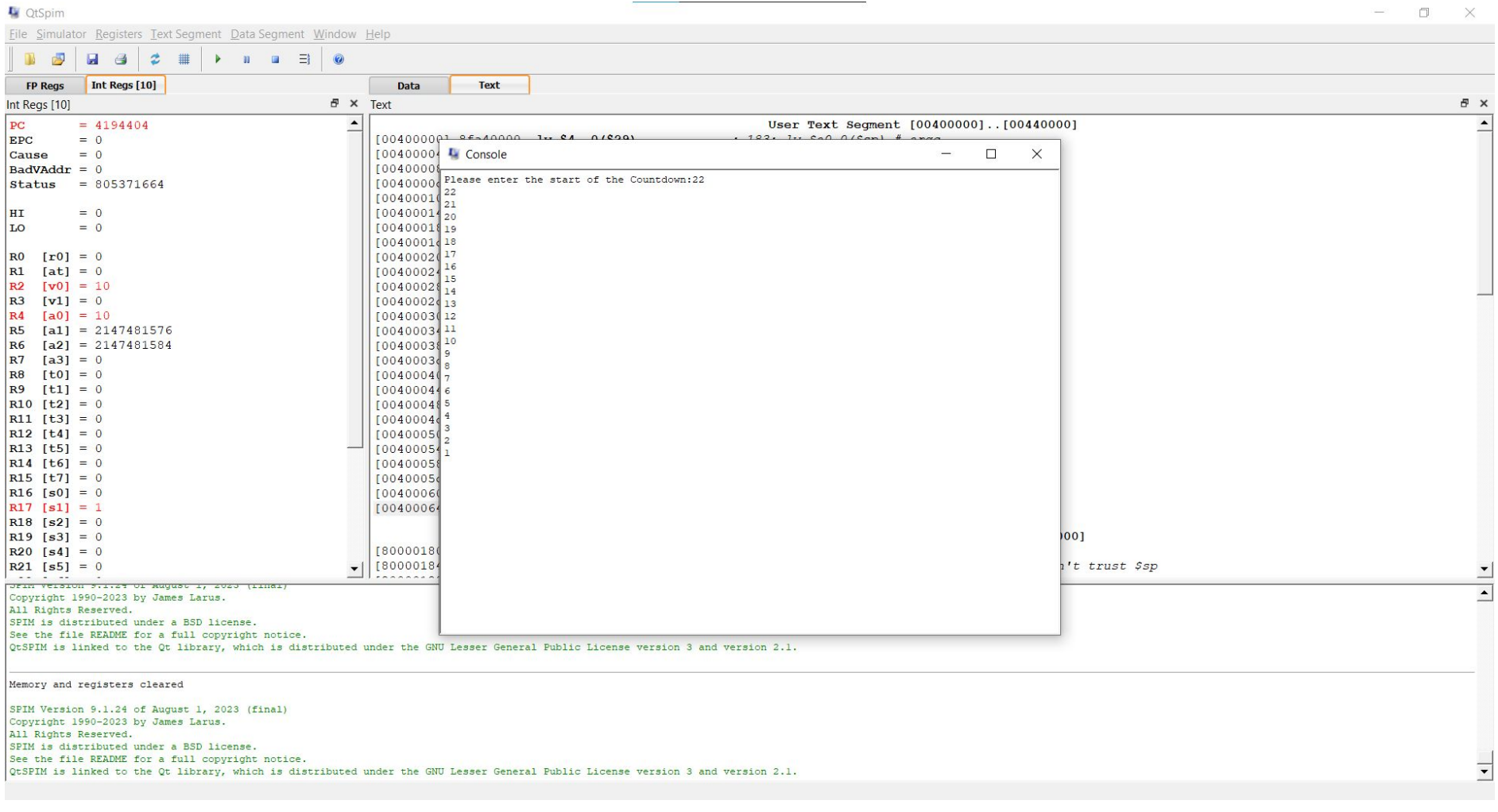

QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Data
Text
Int Regs [10]
PC = 4194640
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 0
R0 [r0] = 0
R1 [at] = 268500992
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 13
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 13
R9 [t1] = 8
R10 [t2] = 0
R11 [t3] = 268697628
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 8
R17 [s1] = 13
R18 [s2] = 21
R19 [s3] = 7
R20 [s4] = 0
R21 [s5] = 0
Text
[004000d8] 000b2021 addu $4, $0, $11 ; 70: move $a0, $t3
Console
Please enter how many fibonacci numbers you want:8
1 number of fibonacci series is stored at memory location 268697600
2 number of fibonacci series is stored at memory location 268697604
3 number of fibonacci series is stored at memory location 268697608
4 number of fibonacci series is stored at memory location 268697612
5 number of fibonacci series is stored at memory location 268697616
6 number of fibonacci series is stored at memory location 268697620
7 number of fibonacci series is stored at memory location 268697624
8 number of fibonacci series is stored at memory location 268697628
fibonacci series: 0,1,1,2,3,5,8,13
Copyright 1990-2023 by James Larus.
All Rights Reserved.
SPIM is distributed under a BSD license.
See the file README for a full copyright notice.
QtSPIM is linked to the Qt library, which is distributed under the GNU Lesser General Public License version 3 and version 2.1.
Memory and registers cleared
SPIM Version 9.1.24 of August 1, 2023 (final)
Copyright 1990-2023 by James Larus.
All Rights Reserved.
SPIM is distributed under a BSD license.
See the file README for a full copyright notice.
QtSPIM is linked to the Qt library, which is distributed under the GNU Lesser General Public License version 3 and version 2.1.


QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194640
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 0
R0 [r0] = 0
R1 [at] = 268500992
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 0
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 0
R9 [t1] = 1
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 1
R17 [s1] = 0
R18 [s2] = 1
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
[004000d8] 000b2021 addu $4, $0, $11 ; 70: move $a0, $t3
Console
Please enter how many fibonacci numbers you want:1
1 number of fibonacci series is stored at memory location 268697600
fibonacci series: 0
Copyright 1990-2023 by James Larus.
All Rights Reserved.
SPIM is distributed under a BSD license.
See the file README for a full copyright notice.
QtSPIM is linked to the Qt library, which is distributed under the GNU Lesser General Public License version 3 and version 2.1.
Memory and registers cleared
SPIM Version 9.1.24 of August 1, 2023 (final)
Copyright 1990-2023 by James Larus.
All Rights Reserved.
SPIM is distributed under a BSD license.
See the file README for a full copyright notice.
QtSPIM is linked to the Qt library, which is distributed under the GNU Lesser General Public License version 3 and version 2.1.
Input Switch Notification
Input language switching
Typing Left Alt + Shift changes your input language. You can turn this feature off or change your hot key sequence by selecting Customize.
Customize
Dismiss



Error handling for non-positive inputs.


QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194404
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 0
R0 [r0] = 0
R1 [at] = 0
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 10
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 0
R9 [t1] = 0
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 0
R17 [s1] = 1
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
User Text Segment [00400000]..[0
[00400000] 8fa40000 lw $4, 0($29) ; 183: lw $a0 0($sp) # argc
[00400004] 27a50004 addiu $5, $29, 4 ; 184: addiu $a1 $sp 4 # argv
[00400008] 24a60004 addiu $6, $5, 4 ; 185: addiu $a2 $a1 4 # envp
[0040000c] 00041080 sll $2, $4, 2 ; 186: sll $v0 $a0 2
[00400010] 00c23021 addu $6, $6, $2 ; 187: addu $a2 $a2 $v0
[00400014] 0c100009 jal 0x00400024 [main] ; 188: jal main
[00400018] 00000000 nop ; 189: nop
[0040001c] 3402000a ori $2, $0, 10 ; 191: li $v0 10
[00400020] 0000000c syscall ; 192: syscall # syscall 10 (exit)
[00400024] 34020004 ori $2, $0, 4 ; 8: li $v0, 4
[00400028] 3c041001 lui $4, 4097 [ent] ; 9: la $a0, ent
[0040002c] 0000000c syscall ; 10: syscall
[00400030] 34020005 ori $2, $0, 5 ; 12: li $v0, 5
[00400034] 0000000c syscall ; 13: syscall
[00400038] 00028021 addu $16, $0, $2 ; 15: move $s0, $v0
[0040003c] 34110001 ori $17, $0, 1 ; 17: li $s1, 1
[00400040] 34020001 ori $2, $0, 1 ; 20: li $v0, 1
[00400044] 00102021 addu $4, $0, $16 ; 21: move $a0, $s0
[00400048] 0000000c syscall ; 22: syscall
[0040004c] 3402000b ori $2, $0, 11 ; 23: li $v0, 11
[00400050] 3404000a ori $4, $0, 10 ; 24: li $a0, 10
[00400054] 0000000c syscall ; 25: syscall
[00400058] 2210ffff addi $16, $16, -1 ; 26: addi $s0, $s0, -1
[0040005c] 1600fff9 bne $16, $0, -28 [loop-0x0040005c]
[00400060] 3402000a ori $2, $0, 10 ; 28: li $v0, 10
[00400064] 0000000c syscall ; 29: syscall
Kernel Text Segment [80000000]..[
[80000180] 0001d821 addu $27, $0, $1 ; 90: move $k1 $at # Save $at
[80000184] 3c019000 lui $1, -28672 ; 92: sw $v0 s1 # Not re-entrant and
Copyright 1990-2023 by James Larus.
All Rights Reserved.
SPIM is distributed under a BSD license.
See the file README for a full copyright notice.
QtSPIM is linked to the Qt library, which is distributed under the GNU Lesser General Public License version 3 and version 2.1.
Memory and registers cleared
SPIM Version 9.1.24 of August 1, 2023 (final)
Copyright 1990-2023 by James Larus.
All Rights Reserved.
SPIM is distributed under a BSD license.
See the file README for a full copyright notice.
QtSPIM is linked to the Qt library, which is distributed under the GNU Lesser General Public License version 3 and version 2.1.
Console
Please enter the start of the Countdown:67
67
66
65
64
63
62
61
60
59
58
57
56
55
54
53
52
51
50
49
48
47
46
45
44
43
42
41
40
39
38
37
36
35
34
33
32
31
30
29
28
27
26
25
24
23
22
21
20
19
18
17
16
15
14
13
12
11
10
9


QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194404
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 0
R0 [r0] = 0
R1 [at] = 0
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 10
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 0
R9 [t1] = 0
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 0
R17 [s1] = 1
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
User Text Segment [00400000]..[00440000]
Console
Please enter the start of the Countdown:22
22
21
20
19
18
17
16
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
Copyright 1990-2023 by James Larus.
All Rights Reserved.
SPIM is distributed under a BSD license.
See the file README for a full copyright notice.
QtSPIM is linked to the Qt library, which is distributed under the GNU Lesser General Public License version 3 and version 2.1.
Memory and registers cleared
SPIM Version 9.1.24 of August 1, 2023 (final)
Copyright 1990-2023 by James Larus.
All Rights Reserved.
SPIM is distributed under a BSD license.
See the file README for a full copyright notice.
QtSPIM is linked to the Qt library, which is distributed under the GNU Lesser General Public License version 3 and version 2.1.

QtSpim

File Simulator Registers Text Segment Data Segment Window Help

FP Regs | Int Regs [10] | Data | Text

Int Regs [10]

```
PC       = 4194416
EPC      = 0
Cause    = 0
BadVAddr = 0
Status   = 805371664

HI       = 0
LO       = 0

R0  [r0] = 0
R1  [at] = 0
R2  [v0] = 10
R3  [v1] = 0
R4  [a0] = 1
R5  [a1] = 2147481576
R6  [a2] = 2147481584
R7  [a3] = 0
R8  [t0] = 100
R9  [t1] = 0
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 78
R17 [s1] = 0
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
```

Text

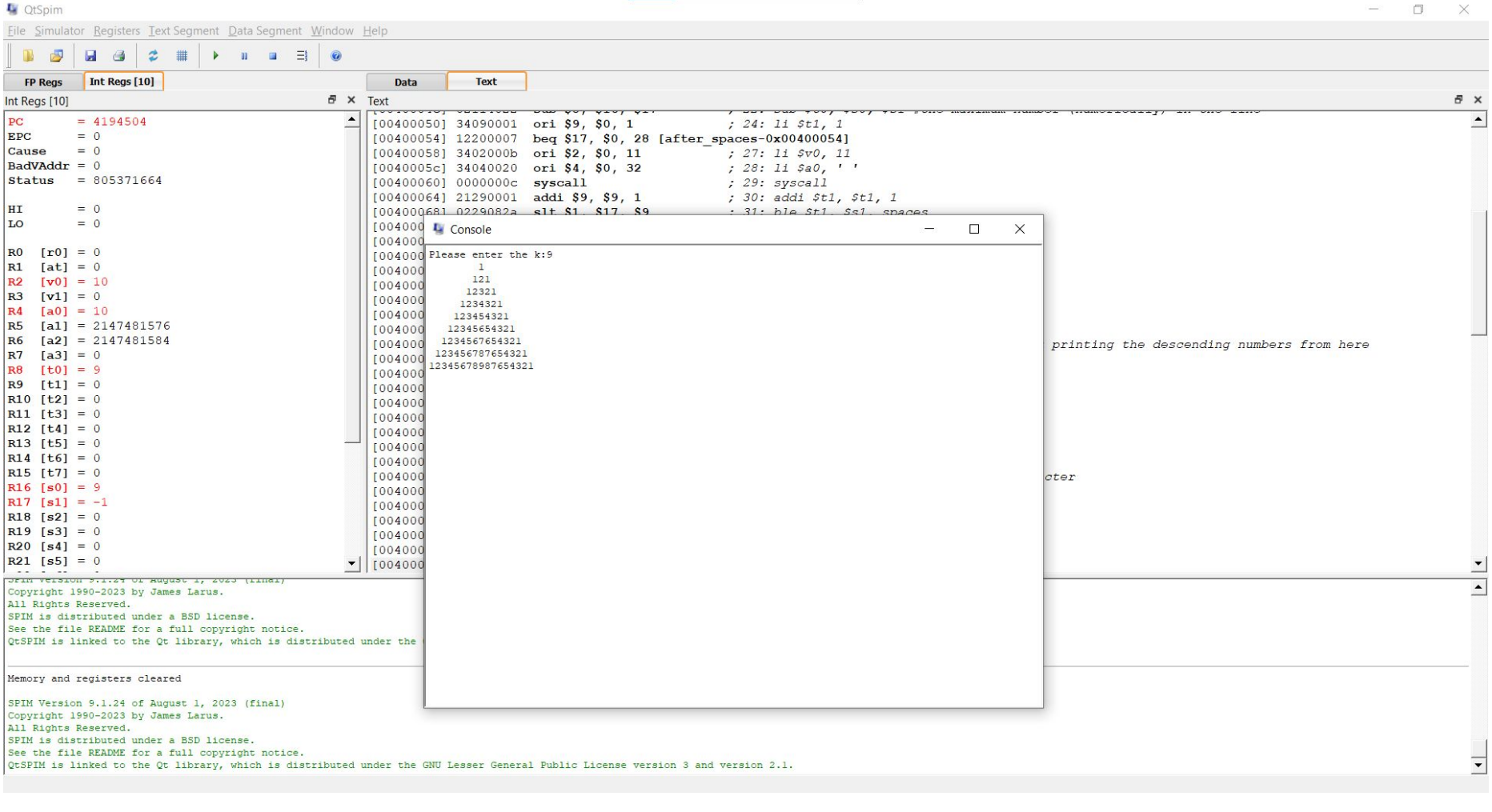
```
                                User Text Segment [00400000]..[00440000]
[00400000] 8fa40000  lw $4, 0($29)            ; 183: lw $a0 0($sp) # argc
[00400004] 27a50004  addiu $5, $29, 4         ; 184: addiu $a1 $sp 4 # argv
[00400008] 24a60004  addiu $6, $5, 4          ; 185: addiu $a2 $a1 4 # envp
[0040000c] 00041080  sll $2, $4, 2            ; 186: sll $v0 $a0 2
[00400010] 00c23021  addu $6, $6, $2          ; 187: addu $a2 $a2 $v0
[00400014] 0c100009  jal 0x00400024 [main]    ; 188: jal main
[00400018] 00000000  nop                      ; 189: nop
[0040001c] 3402000a  ori $2, $0, 10           ; 191: li $v0 10
[00400020] 0000000c  syscall                  ; 192: syscall # syscall 10 (exit)
[00400024] 34100001  ori $16, $0, 1           ; 7: li $s0, 1 #J
[00400028] 34110000  ori $17, $0, 0           ; 8: li $s1, 0 #a
[0040002c] 34080014  ori $8, $0, 20           ; 10: li $t0, 20
[00400030] 0111082a  slt $1, $8, $17          ; 13: bgt $s1, $t0, dowhile
[00400034] 14200004  bne $1, $0, 16 [dowhile-0x00400034]
[00400038] 22100001  addi $16, $16, 1         ; 14: addi $s0, $s0, 1
[0040003c] 22310002  addi $17, $17, 2         ; 15: addi $s1, $s1, 2
[00400040] 0810000c  j 0x00400030 [while]     ; 16: j while
[00400044] 34080064  ori $8, $0, 100          ; 19: li $t0, 100
[00400048] 22100001  addi $16, $16, 1         ; 20: addi $s0, $s0, 1
[0040004c] 0208082a  slt $1, $16, $8          ; 21: bge $s0, $t0, subi
[00400050] 10200002  beq $1, $0, 8 [subi-0x00400050]
[00400054] 08100011  j 0x00400044 [dowhile]   ; 22: j dowhile
[00400058] 0011082a  slt $1, $0, $17          ; 25: ble $s1, $0, end
[0040005c] 10200004  beq $1, $0, 16 [end-0x0040005c]
[00400060] 2210ffff  addi $16, $16, -1        ; 26: addi $s0, $s0, -1
[00400064] 2231ffff  addi $17, $17, -1        ; 27: addi $s1, $s1, -1
[00400068] 08100016  j 0x00400058 [subi]      ; 28: j subi
[0040006c] 3402000a  ori $2, $0, 10           ; 31: li $v0, 10
[00400070] 0000000c  syscall                  ; 32: syscall
```

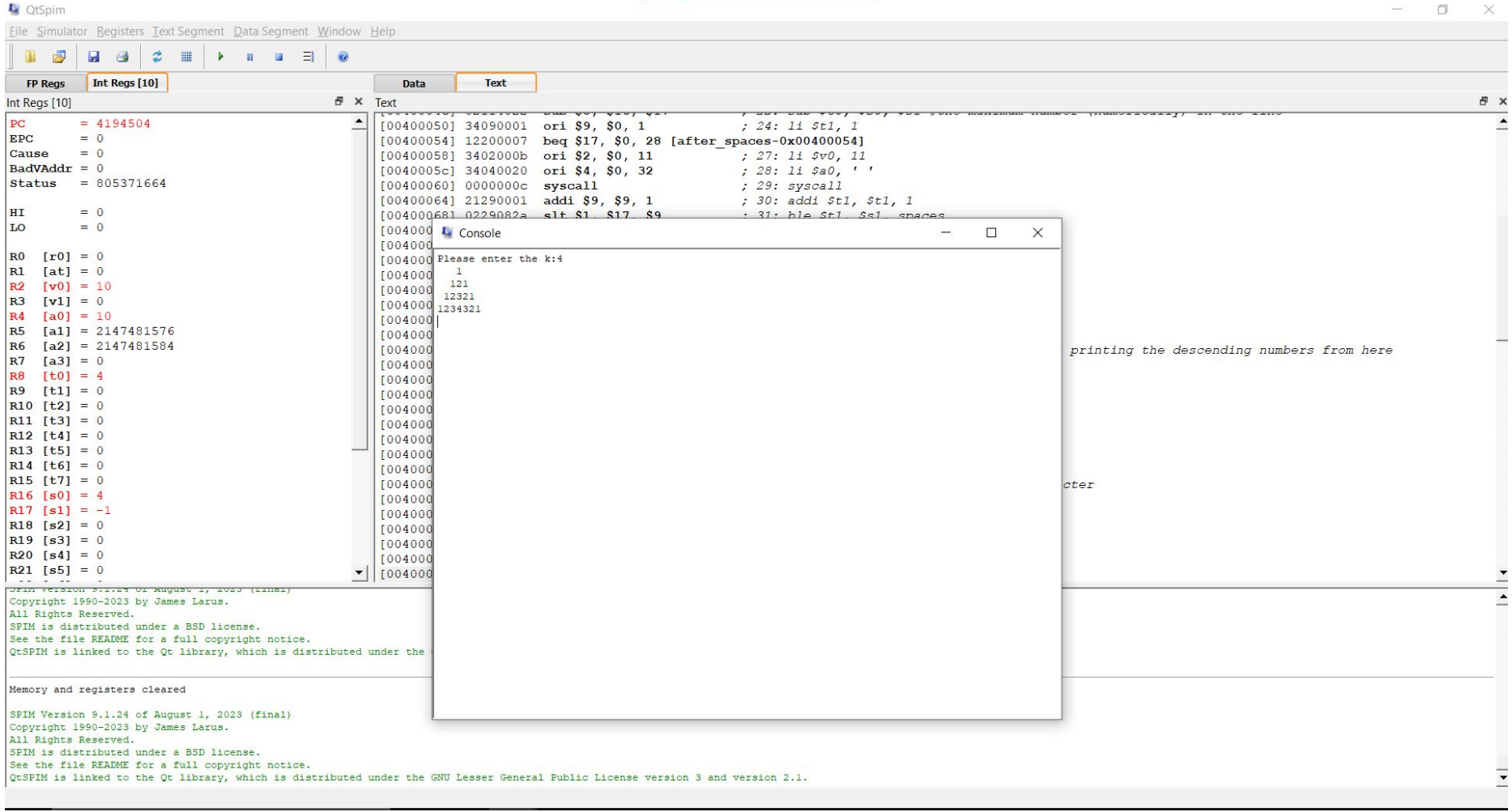
Final values in s0 (J) and s1 (a).

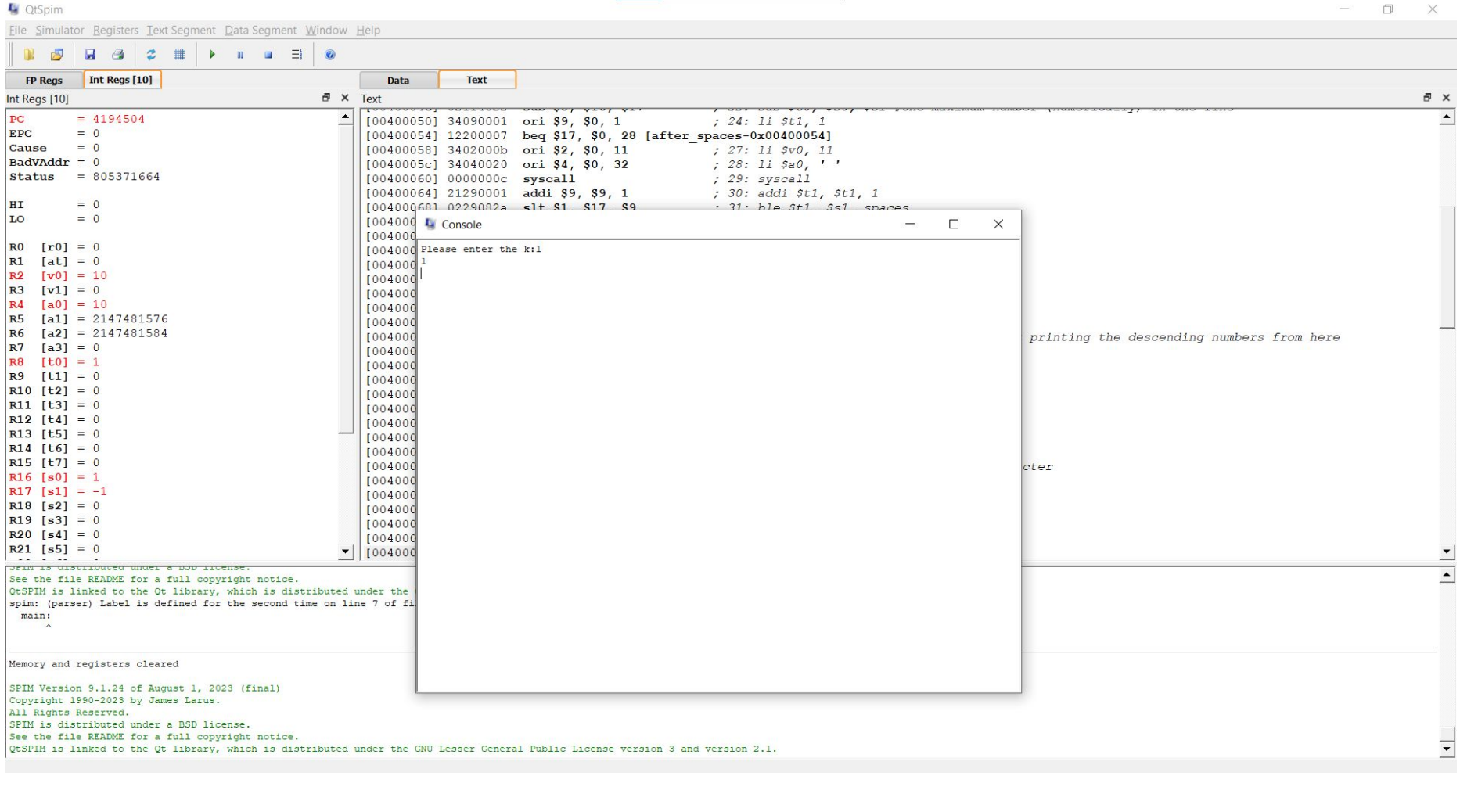
```
See the file README for a full copyright notice.
QtSPIM is linked to the Qt library, which is distributed under the GNU Lesser General Public License version 3 and version 2.1.
spim: (parser) syntax error on line 28 of file C:/Users/valorant_guy/Desktop/Computer)rganizatin/assignment4/a4q3.S
  sub
  ^

Memory and registers cleared

SPIM Version 9.1.24 of August 1, 2023 (final)
Copyright 1990-2023 by James Larus.
All Rights Reserved.
SPIM is distributed under a BSD license.
See the file README for a full copyright notice.
QtSPIM is linked to the Qt library, which is distributed under the GNU Lesser General Public License version 3 and version 2.1.
```


QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194504
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 0
R0 [r0] = 0
R1 [at] = 0
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 10
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 9
R9 [t1] = 0
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 9
R17 [s1] = -1
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
[00400050] 34090001 ori $9, $0, 1 ; 24: li $t1, 1
[00400054] 12200007 beq $17, $0, 28 [after_spaces-0x00400054]
[00400058] 3402000b ori $2, $0, 11 ; 27: li $v0, 11
[0040005c] 34040020 ori $4, $0, 32 ; 28: li $a0, ' '
[00400060] 0000000c syscall ; 29: syscall
[00400064] 21290001 addi $9, $9, 1 ; 30: addi $t1, $t1, 1
printing the descending numbers from here
cter
Console
Please enter the k:9
1
121
12321
1234321
123454321
12345654321
1234567654321
123456787654321
12345678987654321
SPIM Version 9.1.24 of August 1, 2023 (final)
Copyright 1990-2023 by James Larus.
All Rights Reserved.
SPIM is distributed under a BSD license.
See the file README for a full copyright notice.
QtSPIM is linked to the Qt library, which is distributed under the
Memory and registers cleared
SPIM Version 9.1.24 of August 1, 2023 (final)
Copyright 1990-2023 by James Larus.
All Rights Reserved.
SPIM is distributed under a BSD license.
See the file README for a full copyright notice.
QtSPIM is linked to the Qt library, which is distributed under the GNU Lesser General Public License version 3 and version 2.1.


QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194504
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 0
R0 [r0] = 0
R1 [at] = 0
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 10
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 4
R9 [t1] = 0
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 4
R17 [s1] = -1
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
[00400050] 34090001 ori $9, $0, 1 ; 24: li $t1, 1
[00400054] 12200007 beq $17, $0, 28 [after_spaces-0x00400054]
[00400058] 3402000b ori $2, $0, 11 ; 27: li $v0, 11
[0040005c] 34040020 ori $4, $0, 32 ; 28: li $a0, ' '
[00400060] 0000000c syscall ; 29: syscall
[00400064] 21290001 addi $9, $9, 1 ; 30: addi $t1, $t1, 1
printing the descending numbers from here
cter
Console
Please enter the k:4
1
121
12321
1234321
Copyright 1990-2023 by James Larus.
All Rights Reserved.
SPIM is distributed under a BSD license.
See the file README for a full copyright notice.
QtSPIM is linked to the Qt library, which is distributed under the
Memory and registers cleared
SPIM Version 9.1.24 of August 1, 2023 (final)
Copyright 1990-2023 by James Larus.
All Rights Reserved.
SPIM is distributed under a BSD license.
See the file README for a full copyright notice.
QtSPIM is linked to the Qt library, which is distributed under the GNU Lesser General Public License version 3 and version 2.1.


QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194504
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 0
R0 [r0] = 0
R1 [at] = 0
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 10
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 1
R9 [t1] = 0
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 1
R17 [s1] = -1
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
[00400050] 34090001 ori $9, $0, 1 ; 24: li $t1, 1
[00400054] 12200007 beq $17, $0, 28 [after_spaces-0x00400054]
[00400058] 3402000b ori $2, $0, 11 ; 27: li $v0, 11
[0040005c] 34040020 ori $4, $0, 32 ; 28: li $a0, ' '
[00400060] 0000000c syscall ; 29: syscall
[00400064] 21290001 addi $9, $9, 1 ; 30: addi $t1, $t1, 1
printing the descending numbers from here
Console
Please enter the k:1
1
SPIM is distributed under a BSD license.
See the file README for a full copyright notice.
QtSPIM is linked to the Qt library, which is distributed under the
spim: (parser) Label is defined for the second time on line 7 of fi
main:
^
Memory and registers cleared
SPIM Version 9.1.24 of August 1, 2023 (final)
Copyright 1990-2023 by James Larus.
All Rights Reserved.
SPIM is distributed under a BSD license.
See the file README for a full copyright notice.
QtSPIM is linked to the Qt library, which is distributed under the GNU Lesser General Public License version 3 and version 2.1.